

Notes on Linear Algebra (II)

1 Jordan Canonical Form

Definition 1.1

A Jordan block $J_{\lambda,k}$ of eigenvalue λ and index/size k is a $k \times k$ upper triangular matrix such that $\dots$

Theorem 1.1

Let $J = J_{\lambda,k} \in M_k(\mathbb{C})$, then:

1. $p_J(x) = (\lambda - x)^k$.
2. $a_J(\lambda) = k$, and $g_J(\lambda) = 1$.
3. For $1 \leq i \leq k$, $\text{rank}((J - \lambda I_k)^i) = k - i$ and $\text{nullity}((J - \lambda I_k)^i) = i$.
4. $\text{nullity}((J - \lambda I_k)^i) - \text{nullity}((J - \lambda I_k)^{i-1}) = \begin{cases} 1, & \text{if } 1 \leq i \leq k \\ 0, & \text{if } i \geq k + 1 \end{cases}$.
5. $m_J(x) = (x - \lambda)^k$.

Proof.

1,2 is trivial by inspection. For 3 and 4, let $N_k = J - \lambda I_k$. The key observation is that the diagonal of 1's is shifted by one unit to the right after multiplying N_k itself. For 5, since $N_k^i \neq \mathbf{O}$ for $i < k$, the minimal polynomial of J must be the characteristic polynomial of J . ■

Remark. 2 and 5 shows that J is diagonalizable only when J is of size 1.

Definition 1.2

Direct sum of matrices Let $A \in M_m(F)$, $B \in M_n(F)$, then the direct sum $A \oplus B$ is a block diagonal matrix in $M_{m+n}(F)$ such that

$$A \oplus B = \begin{pmatrix} A & \mathbf{O} \\ \mathbf{O} & B \end{pmatrix}$$

Exercise 1.1

Write down the definition of a direct sum of two subspaces. They are two different concepts.

Theorem 1.2

Let $A \in M_m(F)$, $B \in M_n(F)$, then:

1. $\text{rank}(A \oplus B) = \text{rank}(A) + \text{rank}(B)$
2. $\text{nullity}(A \oplus B) = \text{nullity}(A) + \text{nullity}(B)$
3. $p_{A \oplus B}(x) = p_A(x) \cdot p_B(x)$
4. $a_{A \oplus B}(\lambda) = a_A(\lambda) + a_B(\lambda)$
5. $g_{A \oplus B}(\lambda) = g_A(\lambda) + g_B(\lambda)$
6. $m_{A \oplus B}(x) = \text{lcm}\{m_A(x), m_B(x)\}$

Proof.

1–5 are left as an exercise. For 6, observe that for any polynomial $f(x) \in F[x]$, one has $f(A \oplus B) = f(A) \oplus f(B)$. Let $m_{A \oplus B} = f(x)$, $f(A \oplus B) = \mathbf{0}$ by the definition of minimal polynomial, which forces $f(x)$ is a common multiple of $m_A(x)$ and $m_B(x)$. For the "least" part, we provide two methods. Suppose $g(x)$ is another common multiple, we want to show f divides g . Since $m_A(x) \mid g(x)$ and $m_B(x) \mid g(x)$, $g(A \oplus B) = \mathbf{0}$ implies $f \mid g$.

Alternatively, conduct the division $g(x) = f(x) \cdot q(x) + r(x)$. It suffices to show $r(x) \equiv 0$. Suppose, by contradiction, $\deg(r) < \deg(f)$. However, $r(A) = r(B) = \mathbf{0}$ violates the minimality. ■

Definition 1.3 (JCF)

A matrix J is in its Jordan Canonical Form if it is a direct sum of Jordan blocks:

$$J = J_{\lambda_1, k_1} \oplus J_{\lambda_2, k_2} \oplus \cdots \oplus J_{\lambda_n, k_n}.$$

Corollary 1. Any diagonal matrix is exactly in its JCF. . .

2. Write down the $p_J(x)$ and $m_J(x)$ in terms of $p_{J_{\lambda_i, k_i}}(x)$ and $m_{J_{\lambda_i, k_i}}(x)$.

Theorem 1.3

Let J in its JCF and fix $\lambda \in \text{Spec}(J)$, then:

1. # of Jordan blocks of eigenvalue $\lambda = g_J(\lambda)$.

2. # of Jordan blocks of eigenvalue λ of index(size) at least $i = \text{nullity}((J - \lambda I_k)^i) - \text{nullity}((J - \lambda I_k)^{i-1})$.

Proof.

Let $J = J_{\lambda_1, k_1} \oplus J_{\lambda_2, k_2} \oplus \cdots \oplus J_{\lambda_n, k_n}$. Suppose $\lambda_1 = \lambda_2 = \cdots \lambda_l = \lambda$, $l \leq n$, $\lambda \notin \{\lambda_{l+1}, \lambda_{l+2}, \cdots \lambda_n\}$. Let $N_k = J - \lambda I_k$, recall (3) and (4) in theorem (1.1),

1. $g_J(\lambda) = \sum_{i=1}^n \text{nullity}(N_{\lambda_i, k_i}) = \underbrace{1 + 1 + \cdots + 1}_{l \text{ times}} + \underbrace{0 + 0 + \cdots + 0}_{\lambda \neq \lambda_i, i \geq l} = l$, which equals # of Jordan blocks of eigenvalue λ .

2. Because $N_k = N_{k_1} \oplus N_{k_2} \oplus \cdots \oplus N_{k_l} \oplus T$, where T is some direct sum of Jordan blocks of eigenvalue that is not λ . Fix i , $\text{nullity}(N_k^i) - \text{nullity}(N_k^{i-1}) = \sum_{j=1}^l \left(\text{nullity}(N_{k_j}^i) - \text{nullity}(N_{k_j}^{i-1}) \right) = \#\{1 \leq j \leq l : i \leq k_j\}$, which equals # of Jordan blocks of eigenvalue λ of index(size) at least i . ■

Combine the definition of JCF and the theroem (1), we can define the JCF of a general matrix $A \in M_n(\mathbb{C})$.

Definition 1.4 (JCF of a general matrix)

Let $A \in M_n(\mathbb{C})$, the JCF of A is a matrix of J_A s.t.

1. J_A is a direct sum of Jordan blocks.
2. # of Jordan blocks of eigenvalue $\lambda = g_A(\lambda)$.
3. # of Jordan blocks of eigenvalue λ of index(size) at least $i = \text{nullity}((A - \lambda I_k)^i) - \text{nullity}((A - \lambda I_k)^{i-1})$.

Remark. How to find the JCF of A ?

1. Find spectrum of A via $p_A(x)$.
2. Find $g_A(\lambda)$ to determine # of Jordan blocks.
3. Find $r_i = \text{nullity}((A - \lambda I_k)^i) - \text{nullity}((A - \lambda I_k)^{i-1})$ to determine individual size of a Jordan block.

Let $T : V \rightarrow V$ be a linear operator over a finite-dimensional complex vector space V . In the context of diagonalization, we aim to find an eigenbasis η such that $[T]_\eta^\eta$ is a diagonal matrix. Likewise, our goal now is to find a basis γ such that $[T]_\gamma^\gamma$ is in its JCF. We first look at a part of matrix corresponding to an eigenvalue λ . Suppose there

exists such a part of basis $\beta = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$, and we put each elements by columns to form a matrix P , then we have:

$$[T]_B^B P = P [T]_\beta^\beta$$

$$T([\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k]) = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k] \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & \cdots & 0 & \lambda \end{pmatrix},$$

where B is the standard basis. Comparing the columns on both sides yields the following system of equations:

$$\begin{array}{ll} T(\mathbf{v}_1) = \lambda \mathbf{v}_1 & (T - \lambda \text{id})(\mathbf{v}_1) = \mathbf{0} \\ T(\mathbf{v}_2) = \mathbf{v}_1 + \lambda \mathbf{v}_2 & (T - \lambda \text{id})(\mathbf{v}_2) = \mathbf{v}_1 \\ T(\mathbf{v}_3) = \mathbf{v}_2 + \lambda \mathbf{v}_3 & \iff (T - \lambda \text{id})(\mathbf{v}_3) = \mathbf{v}_2 \\ \vdots & \vdots \\ T(\mathbf{v}_k) = \mathbf{v}_{k-1} + \lambda \mathbf{v}_k & (T - \lambda \text{id})(\mathbf{v}_k) = \mathbf{v}_{k-1} \end{array}$$

Observe that:

1. $\mathbf{v}_1$ is the only eigenvector among β .
2. Given $\mathbf{v}_k$, we can generate a chain of β by setting $\{\mathbf{v}_k, (T - \lambda \text{id})(\mathbf{v}_k), (T - \lambda \text{id})(\mathbf{v}_{k-1}), \dots, (T - \lambda \text{id})(\mathbf{v}_2)\}$, or equivalently, $\{\mathbf{v}_k, (T - \lambda \text{id})(\mathbf{v}_k), (T - \lambda \text{id})^2(\mathbf{v}_k), \dots, (T - \lambda \text{id})^{k-1}(\mathbf{v}_k)\}$.
3. $(T - \lambda \text{id})^k(\mathbf{v}_k) = (T - \lambda \text{id})(\mathbf{v}_1) = \mathbf{0}$, so $\mathbf{v}_k \in \ker(T - \lambda \text{id})^k \setminus \ker(T - \lambda \text{id})^{k-1} (\because \mathbf{v}_1 \neq \mathbf{0})$.

These motivates the definition of Jordan chain:

Definition 1.5 (Jordan Chain)

Let $T : V \rightarrow V$ be a linear operator and $\lambda \in \text{Spec}(T)$. We say a set of non-zero vectors $\beta = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a Jordan chain of T corresponding to the eigenvalue λ of length k if $\dots$, where $\mathbf{v}_k$ is the end vector and $\mathbf{v}_1$ is the initial vector.